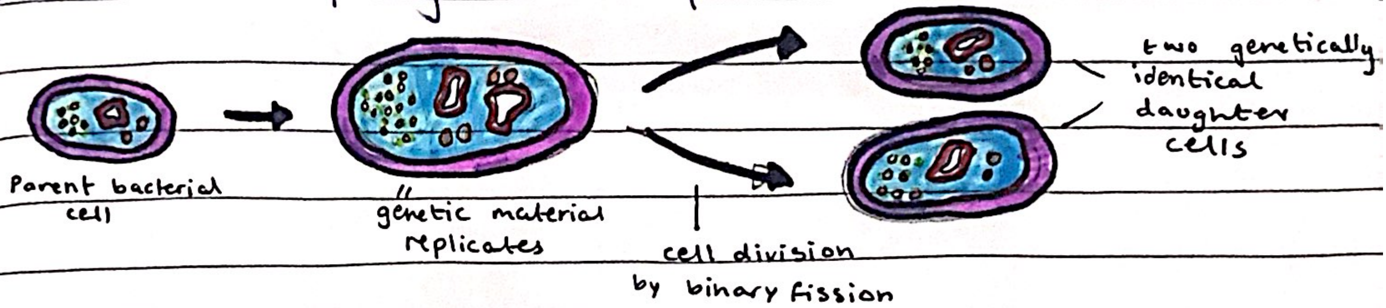
ASEXUAL REPRODUCTION

18/01/23

Asexual reproduction :-

- a process resulting in the production of genetically identical offspring from one parent.



Advantages & Disadvantages of Asexual Reproduction

ADVANTAGES	DISADVANTAGES
POPULATION CAN BE INCREASED RAPIDLY WHEN CONDITIONS ARE RIGHT	LIMITED GENETIC VARIATION IN POPULATION – OFFSPRING ARE GENETICALLY IDENTICAL TO THEIR PARENTS
CAN EXPLOIT SUITABLE ENVIRONMENTS QUICKLY	POPULATION IS VULNERABLE TO CHANGES IN CONDITIONS AND MAY ONLY BE SUITED FOR ONE HABITAT
MORE TIME AND ENERGY EFFICIENT	DISEASE IS LIKELY TO AFFECT THE WHOLE POPULATION AS THERE IS NO GENETIC VARIATION
REPRODUCTION IS COMPLETED MUCH FASTER THAN SEXUAL REPRODUCTION	

- Specifically in **crop plants**, asexual reproduction can be advantageous as it means that a plant that has good characteristics (high yield, disease-resistant, hardy) can be made to reproduce asexually and the **entire crop will show the same characteristics**

SEXUAL REPRODUCTION

18/01/23

Sexual reproduction :-

- a process involving the fusion of the nuclei of two gametes to form a zygote and the production of oppspring that are genetically different from each other
- nuclei of gametes are haploid and the nucleus of a zygote is diploid.

- haploid :- the presence of a single set of chromosomes
- diploid :- the presence of two sets of chromosomes, one from each parent.

fertilisation is the fusion of the nuclei of gametes (the union of a human egg & sperm)

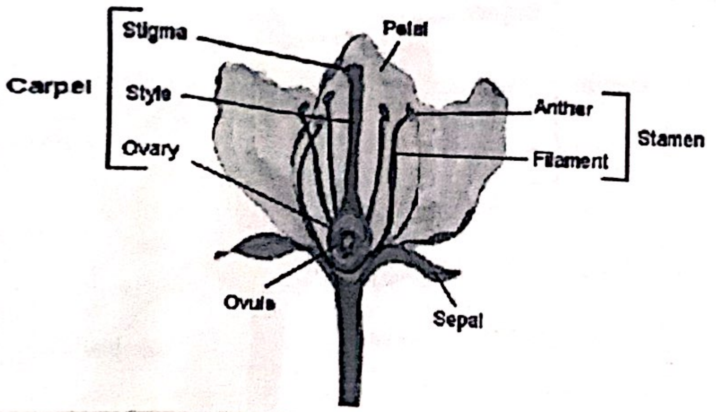
Advantages & Disadvantages of Sexual Reproduction

ADVANTAGES	DISADVANTAGES
INCREASES GENETIC VARIATION	TAKES TIME AND ENERGY TO FIND MATES
THE SPECIES CAN ADAPT TO NEW ENVIRONMENTS DUE TO VARIATION, GIVING THEM A SURVIVAL ADVANTAGE	DIFFICULT FOR ISOLATED MEMBERS OF THE SPECIES TO REPRODUCE
DISEASE IS LESS LIKELY TO AFFECT POPULATION (DUE TO VARIATION)	

- Most crop plants reproduce sexually and this is an advantage as it means **variation is increased** and a genetic variant may be produced which is better able to cope with weather changes, or produces significantly higher yield
- The disadvantage is that the variation may lead to offspring that are **less successful** than the parent plant at growing well or producing a good harvest

SEXUAL REPRODUCTION IN PLANTS

18/01/23



Parts of the flower

STRUCTURE	DESCRIPTION
SEPAL	PROTECTS UNOPENED FLOWER
PETALS	BRIGHTLY COLOURED IN INSECT - POLLINATED FLOWERS TO ATTRACT INSECTS
ANTHER	PRODUCES AND RELEASES THE MALE SEX CELL (POLLEN GRAIN)
STIGMA	TOP OF THE FEMALE PART OF THE FLOWER WHICH COLLECTS POLLEN GRAINS
OVARY	PRODUCES THE FEMALE SEX CELL (OVUM)
OVULE	CONTAINS THE FEMALE SEX CELLS (FOUND INSIDE THE OVARY)

insect-pollinating plant

- in a wind-pollinated plant; the anthers are outside the flower, swinging loose on long filaments to release pollen grains easily & the stigma's are outside the flower, feathery to catch drifting pollen grains

* insect-pollinated flowers produce smaller amounts of large, heavier pollen grains that often contain spikes/hooks on the outside to stick to insects better

wind-pollinated flowers produce larger amounts of small, lightweight pollen grains that are usually smooth.

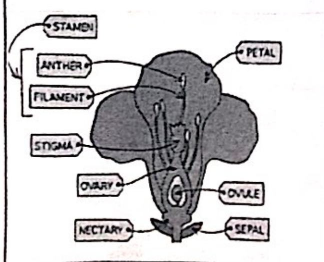
- Pollination is the transfer of pollen grains from an anther to a stigma

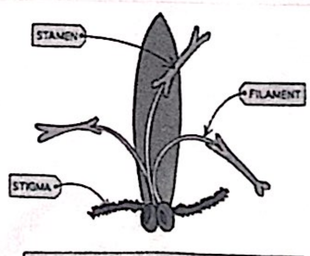
- Cross-pollination is the transfer of pollen grains from the anther of a flower to the stigma of a flower on a different plant of the same species

- Self-pollination is the transfer of pollen grains from the anther of a flower to the stigma of the same or different flower on the same plant

Cross-pollination	Self-pollination
- more variation	- less variation
- has the ability to adapt to changes (in environment)	- doesn't have the ability to adapt to changes
- relies on pollinators	- doesn't rely on pollinators

STRUCTURAL ADAPTATIONS

FEATURE	INSECT-POLLINATED
	 FEATURES OF AN INSECT-POLLINATED FLOWER
PETALS	LARGE AND BRIGHTLY COLOURED TO ATTRACT INSECTS
SCENT AND NECTAR	PRESENT - ENTICES INSECTS TO VISIT THE FLOWER AND PUSH PAST STAMEN TO GET TO NECTAR
NUMBER OF POLLEN GRAINS	MODERATE - INSECTS TRANSFER POLLEN GRAINS EFFICIENTLY WITH A HIGH CHANCE OF SUCCESSFUL POLLINATION
POLLEN GRAINS	LARGER, STICKY AND / OR SPIKY TO ATTACH TO INSECTS AND BE CARRIED AWAY
ANTHERS	INSIDE FLOWER, STIFF AND FIRMLY ATTACHED TO BRUSH AGAINST INSECTS
STIGMA	INSIDE FLOWER, STICKY SO POLLEN GRAINS STICK TO IT WHEN AN INSECT BRUSHES PAST

FEATURE	WIND-POLLINATED
	 FEATURES OF A WIND-POLLINATED FLOWER
PETALS	SMALL AND DULL, OFTEN GREEN OR BROWN IN COLOUR
SCENT AND NECTAR	ABSENT - NO NEED TO WASTE ENERGY PRODUCING THESE AS NO NEED TO ATTRACT INSECTS
NUMBER OF POLLEN GRAINS	LARGE AMOUNTS - MOST POLLEN GRAINS ARE NOT TRANSFERRED TO ANOTHER FLOWER SO THE MORE PRODUCED, THE BETTER THE CHANCE OF SOME SUCCESSFUL POLLINATION OCCURRING
POLLEN GRAINS	SMOOTH, SMALL AND LIGHT SO THEY ARE EASILY BLOWN BY THE WIND
ANTHERS	OUTSIDE FLOWER, SWINGING LOOSE ON LONG FILAMENTS TO RELEASE POLLEN GRAINS EASILY
STIGMA	OUTSIDE FLOWER, FEATHERY TO CATCH DRIFTING POLLEN GRAINS

* Fertilisation occurs when a pollen nucleus fuses with a nucleus of/in an ovule.

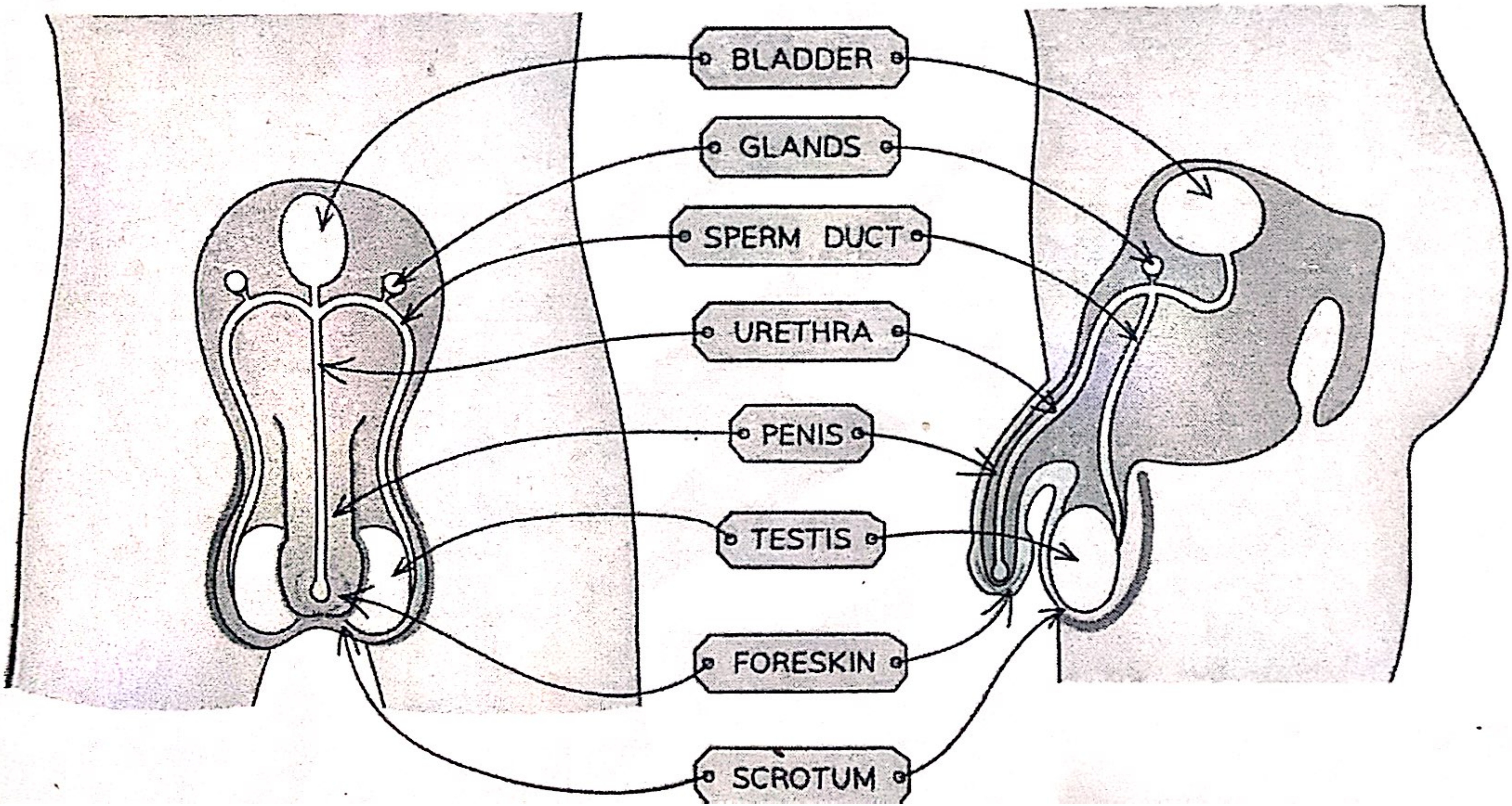
⇒ fertilisation
- the pollen has no 'tail' to swim to the ovary of a plant, in order to reach the 'female' nucleus in the ovary, it has to grow a pollen tube
  └ this only only happens if the pollen grain lands on the right kind of stigma (i.e the same species)
- the nucleus of the pollen grain slips down the tube as it grows down the style, towards ovary,
  └ the ovary contains one or more ovules which contain an ovum (gamete) that a male pollen nucleus can fuse with.
- once the nuclei have joined together, that ovule has been fertilised and a zygote has been formed
  └ a zygote will start to divide and eventually form a seed within the ovule
- different plants have different numbers of ovules, which is why different fruits (developed in ovaries) have different numbers of seeds (developed from the ovules)

Germination:
- the start of growth in the seed

- 3 factors
  └ water; allows seed to swell up & enzymes to start working (in embryo)
  └ oxygen: so that energy can be released for germination
  └ warmth: germination improves as temp. rises (to max.) as reactions taking place are controlled by enzymes

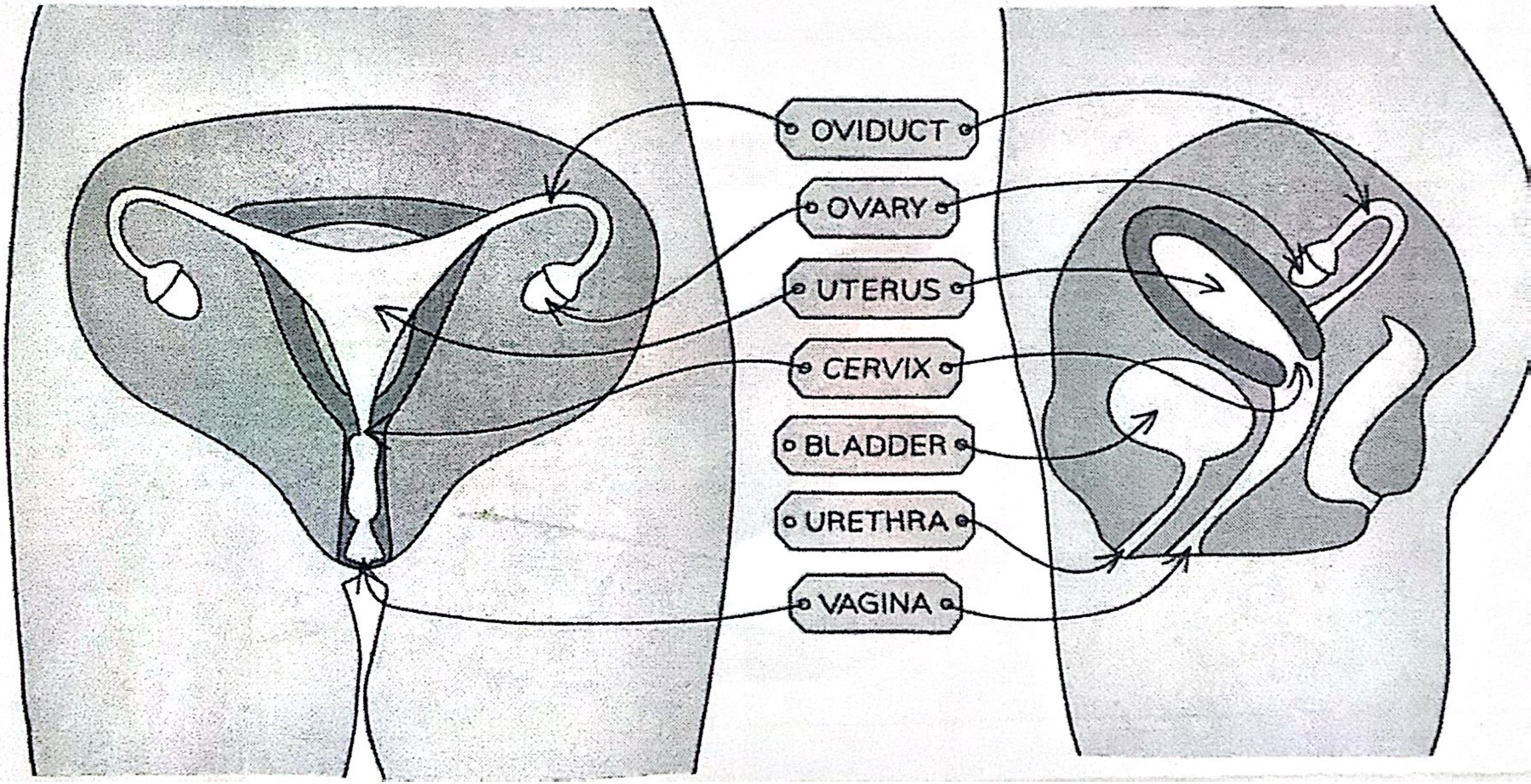
* CO_2 is not necessary

The Male Reproduction System



STRUCTURE	FUNCTION
PROSTATE GLAND	PRODUCES FLUID CALLED SEMEN THAT PROVIDE SPERM CELLS WITH NUTRIENTS
SPERM DUCT	SPERM PASSES THROUGH THE SPERM DUCT TO BE MIXED WITH FLUIDS PRODUCED BY THE GLANDS BEFORE BEING PASSED INTO THE URETHRA FOR EJACULATION
URETHRA	TUBE RUNNING DOWN THE CENTRE OF THE PENIS THAT CAN CARRY OUT URINE OR SEMEN, A RING OF MUSCLE IN THE URETHRA PREVENTS THE URINE AND SEMEN FROM MIXING
TESTIS	CONTAINED IN A BAG OF SKIN (SCROTUM) AND PRODUCES SPERM (MALE GAMETE) AND TESTOSTERONE (HORMONE)
SCROTUM	SAC SUPPORTING THE TESTES OUTSIDE THE BODY TO ENSURE SPERM ARE KEPT AT TEMPERATURE SLIGHTLY LOWER THAN BODY TEMPERATURE
PENIS	PASSES URINE OUT OF THE BODY FROM THE BLADDER AND ALLOWS SEMEN TO PASS INTO THE VAGINA OF A WOMAN DURING SEXUAL INTERCOURSE

The Female Reproduction System



STRUCTURE	FUNCTION
OVIDUCT	CONNECTS THE OVARY TO THE UTERUS AND IS LINED WITH CILIATED CELLS TO PUSH THE RELEASED OVUM DOWN IT. FERTILISATION OCCURS HERE
OVARY	CONTAINS OVA (FEMALE GAMETES) WHICH WILL MATURE AND DEVELOP WHEN HORMONES ARE RELEASED
UTERUS	MUSCULAR BAG WITH A SOFT LINING WHERE THE FERTILISED EGG (ZYGOTE) WILL BE IMPLANTED TO DEVELOP INTO A FOETUS
CERVIX	RING OF MUSCLE AT THE LOWER END OF THE UTERUS TO KEEP THE DEVELOPING FOETUS IN PLACE DURING PREGNANCY
VAGINA	MUSCULAR TUBE THAT LEADS TO THE INSIDE OF THE WOMAN'S BODY, WHERE THE MALE'S PENIS WILL ENTER DURING SEXUAL INTERCOURSE AND SPERM ARE DEPOSITED

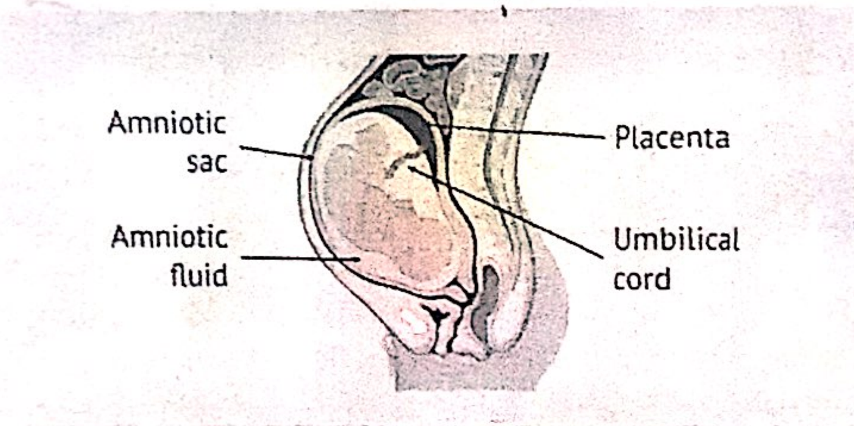
* fertilisation is the fusion of the nuclei from a male gamete (sperm) and a female gamete (egg cell)

GAMETE	ADAPTIVE FEATURE	REASON
SPERM	HAS A FLAGELLUM (TAIL)	ENABLES IT TO SWIM TO THE EGG
	CONTAINS ENZYMES IN THE HEAD REGION (ACROSOME)	TO DIGEST THROUGH THE JELLY COAT AND CELL MEMBRANE OF AN EGG CELL WHEN IT MEETS ONE
	CONTAINS MANY MITOCHONDRIA	PROVIDE ENERGY FROM RESPIRATION SO THAT THE FLAGELLUM CAN MOVE BACK AND FORTH FOR LOCOMOTION
EGG	CYTOPLASM CONTAINING A STORE OF ENERGY	PROVIDES ENERGY FOR THE DIVIDING ZYGOTE AFTER FERTILISATION
	JELLY LIKE COATING THAT CHANGES AFTER FERTILISATION	FORMS AN IMPENETRABLE BARRIER AFTER FERTILISATION TO PREVENT OTHER SPERM NUCLEI ENTERING THE EGG CELL

Comparison

	SPERM	EGG
SIZE	VERY SMALL (45 μm)	LARGE (0.15 mm)
STRUCTURE	HEAD REGION AND FLAGELLUM, MANY STRUCTURAL ADAPTATIONS	ROUND CELL WITH FEW STRUCTURAL ADAPTATIONS, COVERED IN A JELLY COATING
MOTILITY	CAPABLE OF LOCOMOTION	NOT CAPABLE OF LOCOMOTION
NUMBERS	PRODUCED EVERY DAY IN HUGE NUMBERS (AROUND 100 MILLION PER DAY)	THOUSANDS OF IMMATURE EGGS IN EACH OVARY, BUT ONLY ONE RELEASED EACH MONTH

→ in early development, the zygote forms an embryo which is a ball of cells that implants into the lining of the uterus.



* the fetus is surrounded by an amniotic sac which contains amniotic fluid (made from mother's blood plasma)

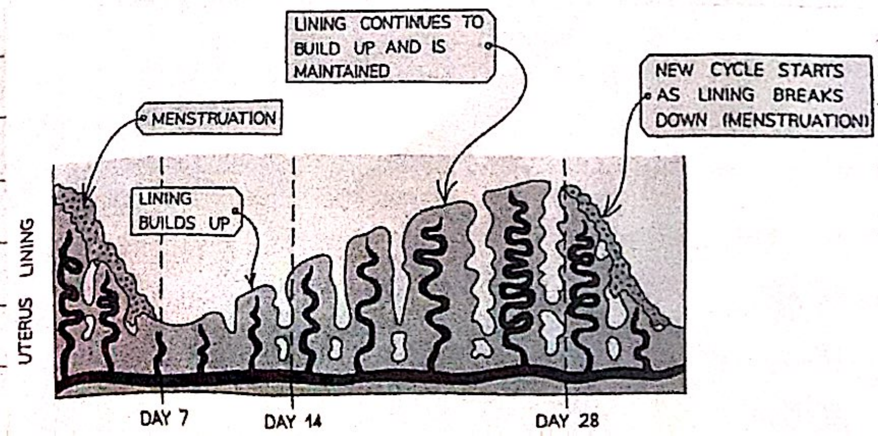
- the fluid cushions the fetus from bumps to the mother's abdomen
- the umblical cord joins the fetus's blood supply to the placenta for exchange of dissolved nutriets, gases & execretory products between the blood of the mother and the blood of the fetus.

* some pathogens and toxins such as nicotine (found in cigarettes) can pass across the placenta and affect the fetus.

→ testosterone (produced in the testes) and oestrogen (produced in the ovaries) both cause the development of secondary sexual characteristics.

testosterone	oestrogen
- growth of penis & testes	- breasts develop
- growth of facial/body hair	- body hair grows
- muscles develop	- menstrual cycle begins
- voice breaks	
- testes start producing sperm	- hips get wider

- average menstrual cycle is 28 days long
  - ovulation (release of an egg) occurs about halfway through the cycle (Day 14) and the egg travels down the oviduct to the uterus.
    - failure to fertilise the egg causes menstruation
      - this is caused by the break down of the thickened lining of the uterus
        - menstruation lasts around 5-7 days
          - after finishing, the uterus starts to thicken again.



oestrogen & progesterone

in the menstrual cycle::

- oestrogen produced in the ovary by the developing follicle
- progesterone produced by the corpus luteum

in pregnancy::

- oestrogen produced in the ovary by the developing follicle
- progesterone produced by the placenta

- FSH (follicle-stimulating hormone) is released by the pituatry gland and causes an egg to start maturing in the ovary. It also stimulates the ovary to start releasing oestrogen. The pituatry gland is stimulated to release LH (luteinising hormone) when oestrogen levels have reached their peak. LH causes an ovulation to occur and also stimulates the ovary to produce progesterone.

Interaction between all four of the menstrual cycle hormones

The pituitary gland produces **FSH** which stimulates the development of a **follicle** in the ovary
An egg develops inside the follicle and the follicle produces the hormone **oestrogen**
Oestrogen causes **growth and repair of the lining of the uterus wall** and **inhibits** production of **FSH**
When oestrogen rises to a high enough level it stimulates the release of **LH** from the pituitary gland which causes **ovulation** (usually around day 14 of the cycle)
The follicle becomes the **corpus luteum** and starts producing **progesterone**
Progesterone **maintains the uterus lining** (the thickness of the uterus wall)
If the ovum is not fertilised, the corpus luteum breaks down and progesterone levels drop
This causes **menstruation**, where the uterus lining breaks down and is removed through the vagina - commonly known as having a period
If pregnancy does occur the corpus luteum **continues to produce progesterone**, preventing the uterus lining from breaking down and **aborting** the pregnancy
It does this until the **placenta** has developed, at which point it starts secreting progesterone **and continues to do so throughout the pregnancy**

SEXUALLY TRANSMITTED INFECTIONS

21/01/23

Sexually transmitted infections:: (STI)

- an infection that is transmitted through sexual contact

- HIV (human immunodeficiency virus) is a pathogen that causes an STI
- HIV infection may lead to AIDS
  - can also be spread via sharing needles with an infected person, blood transfusion with infected blood and from mother to fetus through the placenta and mother to baby via breastfeeding.

Controlling the spread of STI:

- limiting the number of sexual partners an individual has
- not having unprotected sex, but making sure to always use a condom
- getting tested if unprotected sex or sex with multiple partners has ocurred
- raising awareness by education programmes.